span_corruption.py — technical

`src/moe_pruning/data/span_corruption.py · 104 lines · moe_pruning.data.span_corruption`

Purpose

T5 span corruption, deterministic per example.

Formulas

```
num_noise = clamp(round(L * density), 1, L - 1); num_spans = max(round(num_noise / mean_span), 1)
noise and non-noise tokens are each split into num_spans random segments, interleaved non-noise first
```

Step by step

Classes and functions in source order. Signatures, line ranges, raised exceptions and settings are extracted from the code; numbered steps describe what the code does.

`random_spans_noise_mask`

```
def random_spans_noise_mask(length: int, noise_density: float, mean_noise_span_length: float, rng: np.random.Generator) -> np.ndarray
```

lines 18-45

Boolean mask of `length` with ~noise_density noise in spans of mean length.

1. Compute the number of noise tokens and spans (formulas above).
2. `_random_segmentation(n, k)`: shuffle a boolean vector with k-1 True among n-1 positions, prefix a False, cumulative-sum to segment ids, count segment lengths -> k positive lengths summing to n.
3. Interleave non-noise and noise span lengths, cumulative-sum to span starts, mark starts, cumulative-sum again; odd span numbers are noise.
4. Return the first `orig_length` entries (handles length 1 by computing with length 2).

`sentinel_ids`

```
def sentinel_ids(tokenizer: PreTrainedTokenizerBase, count: int=100) -> list[int]
```

lines 48-52 · raises ValueError

Token ids of the T5 sentinels <extra_id_0> ... <extra_id_99>.

`replace_spans_with_sentinels`

```
def replace_spans_with_sentinels(tokens: np.ndarray, mask: np.ndarray, sentinels: list[int]) -> list[int]
```

lines 55-70 · raises ValueError

Replace each masked span by one sentinel (in order) and drop the rest of the span.

1. Walk tokens; at the first token of each masked run emit the next sentinel (`<extra_id_0>`, `<extra_id_1>`, ...); drop the rest of the run; copy unmasked tokens.

`class CorruptedExample`

lines 74-79 · dataclass

One span-corruption example: source, corrupted input, target, noise mask.

Field	Type	Default
`example_id`	int	
`source_ids`	list[int]	
`input_ids`	list[int]	
`labels`	list[int]	
`noise_mask`	list[bool]	

`corrupt_example`

```
def corrupt_example(example_id: int, token_ids: list[int], tokenizer: PreTrainedTokenizerBase,
noise_density: float, mean_noise_span_length: float, seed: int, max_source_length: int,
max_target_length: int) -> CorruptedExample
```

lines 82-104

Corrupt one document deterministically and fit it into the length limits.

1. Start with `length = min(len(doc), MAX_SOURCE_LENGTH - 1)` (room for EOS).
2. RNG = `np.random.default_rng([EVAL_SEED, example_id])`; build mask.
3. Input = sentinel-replaced tokens + EOS; target = tokens with the inverted mask sentinel-replaced + EOS.
4. If the target exceeds `MAX_TARGET_LENGTH`, shorten the document by 8 tokens and retry (same seed) until it fits.

Errors raised

Function	Exception	Message (f-string fields in braces)
`sentinel_ids`	ValueError	Tokenizer has no T5 <extra_id_N> sentinel tokens
`replace_spans_with_sentinels`	ValueError	More noise spans than available sentinel tokens